8 Limitations

Despite that R-Tuning demonstrates remarkable performance in selecting and rejecting questions, there are still limitations to consider. First of all, R-Tuning only possesses the ability to say *I am sure* and *I am unsure*, where the confidence is binary. However, generating a quantitative value to verbally express its confidence for questions will be more accurate. Additionally, we only adopt answer checking and uncertainty quantification to evaluate whether relevant knowledge is within the pre-trained model's parametric knowledge. There are other rigorous methods to evaluate, such as comparing the instruction-tuning datasets with the pre-training datasets. One can follow Kandpal et al. (2023) to identify the relevant knowledge by entity linking pre-training datasets. Due to the high computational cost of the entity linking method, we plan to explore optimization methods to improve efficiency in future work.

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